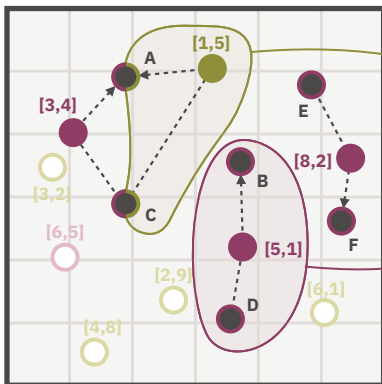


ROUND 75



ComplEx

$$\mathbf{s}_2^- = \text{Re}(\langle \mathbf{C}, [3,4], \bar{\mathbf{A}} \rangle)$$

$$\mathbf{s}^+ = \text{Re}(\langle \mathbf{C}, [1,5], \bar{\mathbf{A}} \rangle)$$

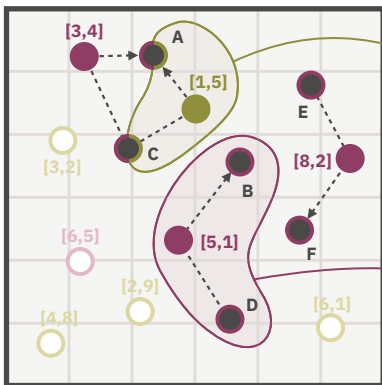
$$\mathbf{s}_1^- = \text{Re}(\langle \mathbf{D}, [5,1], \bar{\mathbf{B}} \rangle)$$

$$\mathbf{s}_3^- = \text{Re}(\langle \mathbf{E}, [8,2], \bar{\mathbf{F}} \rangle)$$

$$L = -\log \frac{e^{\mathbf{s}^+}}{e^{\mathbf{s}^+} + e^{\mathbf{s}_1^-} + e^{\mathbf{s}_2^-} + e^{\mathbf{s}_3^-}}$$

InfoNCE

ROUND 80



ComplEx

$$\mathbf{s}^+ = \text{Re}(\langle \mathbf{C}, [1,5], \bar{\mathbf{A}} \rangle)$$

$$\mathbf{s}_2^- = \text{Re}(\langle \mathbf{C}, [3,4], \bar{\mathbf{A}} \rangle)$$

$$\mathbf{s}_1^- = \text{Re}(\langle \mathbf{D}, [5,1], \bar{\mathbf{B}} \rangle)$$

$$\mathbf{s}_3^- = \text{Re}(\langle \mathbf{E}, [8,2], \bar{\mathbf{F}} \rangle)$$